

COURSE GUIDE

Course / Section	CS313-Automata Theory and Formal Languages	Credit Units	3
Course Description			
The course introduces some fundamental concepts in automata theory and formal languages including grammar, finite automaton, regular expression, formal languages, pushdown automaton, and Turing machines; as well as the applications of these theories in programming languages, compiler design and construction and software development in general. This course is one of the most important courses in Computer Science. Theoretical foundations governing programming languages, compiler design, designing systems in general are covered in this course.			
Course Learning Outcomes			
CLO1: Explain the importance of the course in relation to the goals and objectives of the program, the college and the university CLO2: Demonstrate knowledge of alphabets and languages CLO3: Design/Construct finite-state machines, regular expressions, context-free grammar, push-down automata and Turing machines for modeling a given language CLO4: Design and implement simple lexical analyzers and parser for simple languages using automata concepts CLO5: Discuss the various applications of computer theory in software development.			
Topics / Modules and Intended Learning Outcomes			
<u>Main Reference:</u> <i>Elements of the Theory of Computation by Lewis and Papadimitriou</i> <u>Topic 1: Alphabets, Languages and Regular Expressions (pages 47-51)</u> ILO1. Define the terms symbol, alphabet and languages ILO2. Determine whether a string is generated by a regular expression ILO3. Construct regular expression to represent a particular language <u>Topic 2: Finite Automata (pages 55 - 67)</u> ILO1. Define a finite automaton ILO2. Determine thru tracing whether a particular string is accepted by a DFA ILO3. Construct a DFA to represent a particular language ILO4. Determine thru tracing whether a particular string is accepted by a NFA ILO5. Construct a NFA to represent a particular language <u>Topic 3: Equivalence of DFA, NFA and Regular Expressions (pages 67-85)</u> ILO1. Convert an NFA to a DFA ILO2. Convert a regular expression to NFA ILO3. Convert a DFA to a regular expression ILO4. Explain and illustrate the relationship among regular expression, DFA and NFA			

Topic 4: Applications and Properties of Finite Automata and Regular Languages (pages 86-90)

ILO1. Prove that languages are regular

ILO2. Prove that languages are not regular

ILO3. Apply the theory of finite automata in the construction of a lexical analyzer and other string processing applications

ILO4. Explain how finite automata may be used as a modeling tool in system development

Topic 5: Context-free Grammar (pages 113-120)

ILO1. Differentiate a regular language from a context-free language

ILO2. Define a context-free grammar (CFG)

ILO3. Determine whether a string is generated by a Context-free grammar

ILO4. Construct a context-free grammar for a particular language

ILO5. Construct a regular grammar for a regular language

Topic 6: Push-down Automata (pages 130-150)

ILO1. Define a push-down automaton (PDA)

ILO2. Determine whether a string is accepted by a push-down automaton

ILO3. Construct a push-down automaton for a particular language

ILO4. Prove by construction that a language is context-free

ILO5. Explain the relationship between CFG and PDA and regular languages

Topic 7: Turing Machines (pages 179 – 200)

ILO1. Construct a Turing machine to compute a function

ILO2. Determine whether a string is accepted by a Turing machine

ILO3. Construct a Turing machine to decide a language

ILO4. Construct a Turing machine to accept a language

ILO5. Prove by construction that a language is Turing decidable

ILO6. Prove by construction that a function is Turing computable

ILO7. Prove by construction that a language is Turing acceptable

MADE4Learners FRAMEWORK

Approach	Hybrid	Face-toFace	Online	/	Approach 5
Technical/Software Requirements	- Hardware: PC / laptop - Software/App: PDF reader, MSTeams - Materials, etc: Wifi connection at home - MS Teams				

Class Guidelines

- I am expecting that you have the maturity and discipline to do self-directed learning.
- Integrity is also very important. I expect that you will answer assessment activities on your own without cheating or copying from your classmates.
- This course is highly mathematical. Thus, there are a lot of analysis and problem solving required. This course also builds on topics from the beginning to the end. It will be hard to acquire learning outcomes of succeeding topics if you have not acquired the learning outcomes of previous topics.
- There will be 4 major exams and long quizzes.
- There will also be machine problems/programming assignments which are aligned to course learning outcome 4.
- **The most important outcome is your ability to design/construct models like DFA, NFA, CFG, PDA, TM etc. Naturally, the assessment will also require you to design/construct DFA, NFA, CFG, PDA, TM etc for modeling languages. To prepare for this, you need to practice.**

- I will do my best to provide you with learning resources.
- MS Teams will be our major official form of communication. Announcements and discussion threads will be posted there. It will also be used for file sharing of ebooks, notes, my video recordings.

GRADING SYSTEM

Midterm Grade	MCS (Midterm Class Standing) – activities, quizzes, machine problem PE – Prelim Exam ME – Comprehensive Midterm Exam $MS \text{ (Midterm Score)} = MCS * 0.40 + PE * 0.20 + ME * 0.40$
Final Grade	FCS (Final Class Standing) – activities, quizzes, machine problem PFE – Prefinal Exam FE - Comprehensive Final Exam $FS \text{ (Final Score)} = MS * 0.5 + FCS * 0.20 + PFE * .10 + FE * 0.20$